

Joydeep Biswas

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Current Appointment

Associate Professor, Computer Science Department, The University of Texas at Austin
Associate Director, Texas Robotics, The University of Texas at Austin

Visiting Professor, Nvidia

Education

2014	Ph.D. in Robotics	Carnegie Mellon University
2010	M.S. in Robotics	Carnegie Mellon University
2008	B.Tech. in Engineering Physics	Indian Institute of Technology, Bombay

Achievements and Awards

2025	Third Place, RoboCup 2025 Humanoid League (Adult Size), <i>UT Austin Villa</i>
2025	Best Paper Runner-up, RSS 2025 Workshop on Resilient Off-road Autonomous Robotics
2024	Best Paper Award, ACM CHI Conference on Human Factors in Computing Systems
2024	JP Morgan Faculty Research Award
2022	1st Place, Benchmark Autonomous Robot Navigation (BARN) Challenge, ICRA 2022
2021	NSF CAREER Award
2019	Student Best Poster Award, Northrop Grumman University Symposium
2019	IJCAI Early Career Spotlight
2019	Amazon Research Award
2018	JP Morgan AI Faculty Research Award
2018	Best Demo Award, AAMAS 2018
2018	5th place, RoboCup 2018 Small Size League, <i>UMass MinuteBots</i> , Faculty Team Leader
2017	Lower Bracket 1st place, RoboCup 2017 Small Size League, <i>UMass MinuteBots</i> , Faculty Team Leader
2015	Siebel Scholar, Class of 2015
2015	1st place, RoboCup 2015 Small Size League, <i>CMDragons</i> , Student Team Leader
2014	2nd place, RoboCup 2014 Small Size League, <i>CMDragons</i> , Student Team Leader
2013	2nd place, RoboCup 2013 Small Size League, <i>CMDragons</i> , Student Team Leader
2010	2nd place, RoboCup 2010 Small Size League, <i>CMDragons</i> , Team Member

Employment History

2023 - Present	Associate Professor	Computer Science Department, University of Texas at Austin, TX, USA
2017 - 2023	Technical Advisor	Consumer Robotics, Amazon Lab 126
2019 - 2023	Assistant Professor	Computer Science Department, University of Texas at Austin, TX, USA
2019 - 2025	Adjunct Assistant Professor	College of Information and Computer Sciences, University of Massachusetts Amherst, MA, USA
2015 - 2019	Assistant Professor	College of Information and Computer Sciences, University of Massachusetts Amherst, MA, USA
2015	Post-Doctoral Fellow	Computer Science Department, Carnegie Mellon University, Pittsburgh, PA, USA
2012	Summer Intern	Google Research, Mountain View, CA, USA
2010	Summer Intern	Intel Research, Pittsburgh, PA, USA

Funding

Federal Funding

NIH Award “*Human-Centered Development of Guide-dog Robots*”

Role: Co-PI.

PI: Donghyun Kim (UMass Amherst).

Co-PIs: Ivan Lee (UMass Amherst), Nicholas Giudice (University of Maine).

Period: October 2025 – September 2028.

NSF Award “*Collaborative Research: SLES: No Bad Surprises: Aligning Agent and Human Norms via Specification Refinements*”

Role: Co-PI.

PI: Sandhya Saisubramanian (Oregon State University).

Co-PIs: Houssam Abbas (Oregon State University), Shlomo Zilberstein (UMass Amherst).

Period: October 2024 – September 2028.

Army Research Laboratories Award “*SARA Sprint 3: Long-Range Offroad Introspective Driving*”

Role: PI.

Period: January 2024 – December 2025.

NSF Award “*FMitF: Track I: Program Synthesis for Robot Learning from Demonstrations*”

Role: Co-PI.

PI: Isil Dillig.

Period: October 2023 – September 2027.

NSF Award “*GCR: Community-Embedded Robotics: Understanding Sociotechnical Interactions with Long-term Autonomous Deployments*”

Role: Co-PI.

PI: Luis Sentis.

Co-PIs: Elliott Hauser, Justin Hart, Keri Stephens.
Period: October 2022 – September 2027.

Army Research Laboratories Award “*Human-Guided Learning of Neuro-Symbolic Mission Execution Policies*”

Role: PI.
Co-PI: Isil Dillig.
Period: September 2021 – January 2023.

NSF Award “*NRT-AI: Convergent, Responsible, and Ethical Artificial Intelligence Training Experience for Roboticists*”

Role: Co-PI.
PI: Junfeng Jiao.
Co-PIs: Luis Sentis, Justin Hart.
Period: September 2021 – August 2026.

NSF Award “*CAREER: Robust Perception and Customization for Long-Term Autonomous Mobile Service Robots*”

Role: PI.
Period: April 2021 – March 2026.

NSF Award “*RI: Medium: Introspective Perception and Planning for Long-Term Autonomy*”

Role: PI.
Co-PI: Shlomo Zilberstein (UMass)
Period: July 2020 – June 2023.

NSF Award “*SHF: Small: Interactive Synthesis and Repair For Robot Programs*”

Role: Co-PI.
PI: Arjun Guha (UMass)
Period: June 2020 – May 2023.

DARPA Award “*Advancing Learning via Probabilistic Causal Analysis for Competency Awareness*”

Role: Co-PI.
PI: Charles River Analytics Co-PI: David Jensen (UMass).
Period: October 2019 – September 2022.

Army Futures Command Robotics Center of Excellence “*Persistent Fully Autonomous Multi-Robot Tactics in Complex Environments*”

Role: Co-PI.
PI: Peter Stone Co-PIs: Luis Sentis, Justin Hart
Period: October 2019 – December 2022.

NSF Award “*S&AS: FND: Reliable Semi-Autonomy with Diminishing Reliance on Humans*”

Role: Co-PI.
PI: Shlomo Zilberstein (UMass)
Period: September 2017 – August 2020.

DARPA Award “*Intelligent Model-Based Adaptation for Mobile Robotics*”

Role: Co-PI.
PI: Jonathan Aldrich (CMU) Co-PIs: David Garlan (CMU), Manuela Veloso (CMU), Christian Kaestner (CMU), Claire Le Gouess (CMU).
Period: November 2015 – November 2019.

Competitive Industry Awards

Amazon Faculty Research Award, 2024

Role: PI.
Collaborators: Roberto Martin-Martin
Period: September 2025 – August 2026.

JP Morgan Faculty Research Award, 2023

Role: PI.
Collaborators: Arjun Guha (Northeastern University)
Period: September 2023 – August 2024.

Northrop Grumman Mission Systems’ Research in Applications for Learning Machines (REALM) Consortium

Role: Co-PI.
PI: Shaoshuai Mou (Purdue) Co-PIs: Daniel A. DeLaurentis (Purdue), Bing Liu (UIC)
Period: January 2019 – December 2021.

JP Morgan AI Research Award, 2019

Role: PI.
Period: September 2019 – August 2020.

Amazon Research Award, 2018

Role: PI.
Period: September 2019 – August 2020.

Teaching Experience

Instructor, CS 340I, Fall 2025: The Essentials of AI for Life and Society

University-wide course, University of Texas at Austin

Instructor, CS 393R, Fall 2020, Fall 2021, Spring 2024: Autonomous Robots
Graduate course, University of Texas at Austin

Instructor, CS 109, Fall 2023: The Essentials of AI for Life and Society
University-wide course, University of Texas at Austin

Instructor, CS 388U, Fall 2023: Planning, Search, and Reasoning Under Uncertainty
Online MS course, University of Texas at Austin

Instructor, CS 378H, Fall 2023: F1/10 Autonomous Driving – Honors
Honors Undergraduate course, University of Texas at Austin

Instructor, CS378/ME379M/ME397/ECE394J/ECE379K, Spring 2023: Connected Autonomous Electric Vehicles
Undergraduate course, University of Texas at Austin

Instructor, CS 393R, Spring 2023: Planning, Search, and Reasoning Under Uncertainty
Graduate course, University of Texas at Austin

Instructor, CS 378H, Spring 2022: F1/10 Autonomous Driving – Honors
Honors Undergraduate course, University of Texas at Austin

Instructor, CS 378F, Spring 2020, Spring 2021: F1/10 Autonomous Driving
Undergraduate course, University of Texas at Austin

Instructor, COMPSCI 220, Fall 2017, Fall 2018: Programming Methodology
Undergraduate course, University of Massachusetts Amherst

Instructor, COMPSCI 403, Fall 2016, Spring 2018 : Introduction To Robotics
Undergraduate course, University of Massachusetts Amherst

Instructor, COMPSCI 603, Spring 2016, Spring 2017, Spring 2019 : Robotics
Graduate course, University of Massachusetts Amherst

Instructor, COMPSCI 691BR, Spring 2017 : Building A Robot Soccer Team
Graduate Seminar, University of Massachusetts Amherst

Invited Talks

Reading the Dirt: Learning What Matters (and What Doesn't) for Off-Road Navigation
Keynote talk, RSS 2025 Workshop on Resilient Off-road Autonomous Robotics, June 2025

Robots Without Boundaries: Embracing the Unseen and Tackling Novel Tasks
Amazon Consumer Robotics Symposium, May 2025

Robots Without Boundaries: Embracing the Unseen and Tackling Novel Tasks
Keynote talk, Texas Regional Robotics Symposium, May 2025

Teaching Robots To “Get It Right”
Invited talk, Oregon State University, January 2025

Teaching Robots To “Get It Right”

Keynote talk, The 37th International FLAIRS Conference, May 2024

From Social- To Everything- Navigation

Workshop on Social Robot Navigation: Advances and Evaluation. IROS 2023, October 2023

Towards Context-Aware Robot Navigation

Workshop on Robotic Perception and Mapping: Frontier Vision & Learning Techniques: Advances and Evaluation. IROS 2023, October 2023

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous

University of Maryland, October 2022

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous

Northeastern University, October 2022

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous

Wellesley College, October 2022

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous

Samsung AI Center, NYC, June 2022

Self-Supervised and User-Supervised Adaptation of Autonomous Robots

JP Morgan AI Research Center, NYC, June 2022

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous

Stanford University / Robotics Seminar, April 2022

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous

University of Southern California / CS Colloquium, April 2022

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous

Brown University / BigAI Talk, April 2022

Motion Control and Visual Representation Learning for High-Speed Off-Road Driving

University of Pennsylvania / FITenth Invited Lecture, April 2022

Particle Filters for Mobile Robot Localization

Wellesley College, February 2022

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous

Nvidia, March 2021

Anticipating and Avoiding Failures Using Introspective Perception and Physics-Informed Program Synthesis

MIT Embodied Intelligence Seminar, February 2021

Building Robots For Long-Term Autonomy, And Keeping Them Autonomous

Yale University, April 2019

The Quest for “Always-On” Autonomous Mobile Robots

IJCAI 2019 Early Career Spotlight Talk, August 2019

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous

ICRA 2018 Workshop: Long-term Autonomy and Deployment of Intelligent Robots in the Real-world, May

2018

Building Robots For Long-Term Autonomy, And Keeping Them Autonomous
Carnegie Mellon University, March 2018

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous
Amazon, November 2017

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous
IROS 2017 Workshop: Assistance and Service Robotics in a Human Environment, September 2017

Autonomous Mobile Robot Perception for Changing Environments
ICRA 2016 Workshop: AI for Long-term Autonomy, May 2016

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous.
University of New Hampshire Robotics Seminar Series, March 2016

The Quest for Robust, Reliable, Autonomous Mobile Robots.
Williams College Computer Science Department Colloquium, November 2015

The Quest for Robust, Reliable, Autonomous Mobile Robots.
Vecna Robotics, September 2015

The Quest for Robust, Reliable, Autonomous Mobile Robots.
University of Minnesota, Computer Science & Engineering, April 2015

Vector Map-Based, Non-Markov Localization for Long-Term Deployment of Autonomous Mobile Robots
Google X, April 2015

The Quest for Robust, Reliable, Autonomous Mobile Robots.
University of Massachusetts Amherst, School of Computer Science, March 2015

The Quest for Robust, Reliable, Autonomous Mobile Robots.
University of Massachusetts Amherst, School of Computer Science, March 2015

Panels

Panel Member, 2025 AAAI Presidential Panel on the Future of AI: <https://aaai.org/about-aaai/presidential-panel-on-the-future-of-ai-research/>

Panel Moderator, Ethics Aware Design of AI
2020 Global Analytics Summit: Ethics in AI, Texas McCombs, November 2020

Discussion Panel, Record of Robotics at CMU Part II, A Live Interview with Manuela Veloso.
CMU Record of Robotics Series, October 2020

Last Mile Autonomous Delivery Systems: A Live Webcast Demonstration, and Panel Discussion
UT Good Systems Webinar, September 2020

The Call for an Accelerated Autonomy – Robotics on the Frontlines of a Crisis
Computing In Our New Normal: A UTCS Webinar, May 2020

Professional Service

Outreach Activities

- Science on Screen Series at Amherst Cinema, Amherst MA, 31 October 2018: Presented a introduction to “Christine” within the scientific context of actual self-driving cars. *The Radical Future of Self-Driving Cars*.
- SciTech Cafe, Northampton MA, 23 January 2017: Presented a scientific talk to a general public audience. “Where am I?” and Other Fundamental Questions Robots Think Long and Hard to Answer
- HolyokeCodes, Holyoke MA, 8–12 July 2019: Co-Organized with Arjun Guha, a week-long robotics workshop for high-school students with state-of-the-art soccer-playing robots that we used to compete with at RoboCup. We covered the basic robot sense-plan-act control cycle, computational geometry, and simple adversarial planning. Students implemented building blocks of increasingly complex robot behaviors, leading up to a robot soccer tournament.

Significant Service Roles

- Local Chair, Conference on Robot Learning (CoRL): 2026
- Associate Program Chair, AAAI Conference on Artificial Intelligence (AAAI): 2026
- Senior Member Track Co-Chair, AAAI Conference on Artificial Intelligence (AAAI): 2025
- Associate Editor, The International Journal of Robotics Research: 2023 – present
- Robotics Area Chair, International Joint Conference on Artificial Intelligence (IJCAI): 2023
- Co-Organizer, Texas Regional Robotics Symposium: April 29, 2022
- Associate Editor, Elsevier Robotics and Autonomous Systems: 2019 – present
- RoboCup Federation Trustee: 2021 – Present
- Diversity and Inclusion Co-Chair, AAAI Conference on Artificial Intelligence (AAAI): 2022, 2023
- RoboCup Executive Committee, Small Size League: 2015 – 2021
- Robot Exhibitions Co-Chair, International Joint Conference on Artificial Intelligence (IJCAI): 2021
- RoboCup Symposium Co-Chair: 2020-2021
- Robotics Track Co-Chair, International Conference on Autonomous Agents and Multiagent Systems (AAMAS): 2019
- Associate Editor, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS): 2016

Senior Program Committee

- AAAI Conference on Artificial Intelligence (AAAI): 2024
- AAAI Conference on Artificial Intelligence (AAAI): 2023
- International Joint Conference on Artificial Intelligence (IJCAI): 2022
- International Conference on Autonomous Agents and Multiagent Systems (AAMAS): 2022
- AAAI Conference on Artificial Intelligence (AAAI): 2020
- International Conference on Autonomous Agents and Multiagent Systems (AAMAS): 2021
- Autonomous Robots and Multirobot Systems Workshop (ARMS): 2016

Program Committee / Reviewer

- Autonomous Robots and Multirobot Systems Workshop (ARMS): 2020
- AAAI Symposium on Educational Advances in Artificial Intelligence: 2021
- RoboCup Symposium: 2015, 2016, 2017, 2018, 2019
- AAAI Undergraduate Consortium: 2021
- IEEE/SICE International Symposium on System Integration (SII): 2019
- Robotics: Science and Systems (RSS): 2015, 2016, 2019
- International Symposium on Multi-Robot and Multi-Agent Systems (MRS): 2019
- International Conference on Autonomous Agents and Multiagent Systems (AAMAS): 2017, 2018
- International Conference on Automated Planning and Scheduling (ICAPS) : 2016, 2018, 2021
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS): 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020
- IEEE International Conference on Robotics and Automation (ICRA): 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021
- IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN): 2010
- IEEE Conference on Human-Robot Interaction (HRI): 2016
- International Joint Conference on Artificial Intelligence (IJCAI): 2016
- International Conference on Autonomous Agents and Multiagent Systems (AAMAS): 2016

Journal Reviewing

- IEEE Robotics and Automation Letters (RA-L): 2017, 2018, 2019, 2020, 2021
- IEEE Robotics and Automation Magazine (IEEE-RAM): 2013, 2014, 2015, 2016, 2019
- IEEE Transactions on Robotics (T-RO): 2015, 2018, 2019, 2020
- International Journal of Robotics Research (IJRR): 2016, 2017
- International Journal of Social Robotics (SORO): 2018, 2019

Grant Reviewing

NSF Panelist: 2016(x2), 2018, 2019, 2020, 2021, 2022

University Service

College Level

- Faculty Hiring Committee for Whole Communities Whole Health Cluster Hires: 2019–2020
- CNS Fall Lab Working Group in response to COVID restrictions: 2020

Department Level

- Texas Robotics Machine Shop Committee: 2020–2022
- Texas Robotics Space Committee: 2020–2022
- UTCS Turing Scholars Admissions Committee: 2020–2022
- UTCS Diversity, Equity, and Inclusion Committee: 2020–2022
- UTCS Graduate Admissions Committee: 2019–2020
- UMass CICS Honors Program Director: 2018–2019
- UMass CICS Undergraduate Course Assistant Program Director: 2018–2019
- UMass CICS Graduate Admissions Committee: 2015–2016
- UMass CICS Student Activities Committee: 2015–2016
- UMass CICS Data Science Faculty Hiring Committee: 2016–2017
- UMass CICS Student Activities Committee: 2016–2017

Advising and Thesis Committees

PostDoctoral Supervisor

- Rohan Chandra, 2022 – present
- Kiarash Rahmani, 2022 – present

PhD Supervisor

- Sarah Etter, UT Austin. 2023–present
- Dongmyeong Lee, UT Austin. 2023–present
- Cheng-Chun Hsu, UT Austin. 2023–present
- Arthur Zhang, UT Austin. 2022–present
- Zichao Hu, UT Austin. 2022–present
- Sadanand Modak, UT Austin. 2022–present
- Noah Patton (Co-advised by Isil Dillig), UT Austin. 2022–present
- Eric Hsiung (Co-advised by Swarat Chaudhuri), UT Austin. 2022–present
- Amanda Adkins, UT Austin. 2020–present
- Emily Pruc, UMass Amherst. 2018–2022
- Sadegh Rabiee, UT Austin. 2016–2022, currently at Amazon Lab126
Introspective Perception for Mobile Robots
- Jarrett Holtz, UT Austin. 2015–2022, currently at Bosch Research
Leveraging Program Synthesis for Robust Long-Term Robot Autonomy via Interactive Learning and Adaptation
- Samer Nashed (Changed advisors in 2019), UMass Amherst. 2015–2019
- Spencer Lane (Changed advisors in 2019), UMass Amherst. 2016–2019
- Alyxander Burns (Changed advisors in 2019), UMass Amherst. 2017–2019

Master’s Thesis Supervisor

- Kavan Sikand, UT Austin. 2019–2022
- David Balaban, UMass Amherst, 2016 – 2018
A Real-Time Solver For Time-Optimal Control Of Omnidirectional Robots with Bounded Acceleration

Undergraduate Honors Thesis Supervisor

- Elvin Yang, UT Austin. 2021–2022
Wait, That Feels Familiar: Learning to Extrapolate Human Preferences for Preference-Aligned Path Planning
Best Honors Thesis Award
- Rahul Menon, UT Austin. 2021–2022
Terrain-Adaptive Global Planning from Local Demonstrations
- Shakeel Samsudeen, UT Austin. 2021–2022
Context-Aware Object SLAM

- Nathaniel Plaxton, UT Austin. 2021–2022
Estimating Kinodynamic Uncertainty Using Learned Gaussian Noise Models
- Michael Satanovski, UT Austin, 2021-2022
An Empirical Evaluation of LIDAR Object Detectors for Autonomous Mobile Robots
- Edward Schneeweiss, UMass Amherst, 2015 – 2019
Joint Perception and Planning for Obstacle Avoidance over Non-Planar Terrain
- Kyle Vedder, UMass Amherst, 2015 – 2019
X: Anytime Multiagent Path Planning With Bounded Search*
- George Larionov, UMass Amherst, 2015 – 2016
Human-robot Interaction: Integrating Speech Recognition with a Mobile Robot System

PhD Committee Member

- Connor Basich, UMass Amherst. Supervisor: Shlomo Zilberstein
- Minkyu Kim, UT Austin. Supervisor: Luis Sentis
- Abhinav Verma, UT Austin. Supervisor: Swarat Chaudhuri
- Kyle Hollins Wray, UMass Amherst. Supervisor: Shlomo Zilberstein
- Justin Svegliato, UMass Amherst. Supervisor: Shlomo Zilberstein
- Tiffany Liu, UMass Amherst. Supervisor: Roderic Grupen
- Takeshi Takahashi, UMass Amherst. Supervisor: Roderic Grupen
- Mike Lanighan, UMass Amherst. Supervisor: Roderic Grupen
- Keen Sung, UMass Amherst. Supervisor: Brian Levine
- (Thesis Opponent) ¹, Nils Bore, KTH. Supervisor: John Folkesson

Undergraduate Honors Thesis Committee Member

- Stefan Kussmaul, UMass Amherst. Supervisor: Roderic Grupen
- Karl Schmeckpepper, UMass Amherst. Supervisor: Roderic Grupen

¹ A PhD thesis dissertation in the Swedish doctoral system is formally presented by an external examiner, called the *thesis opponent*. A thesis opponent places the work of the PhD thesis in context with the state of the art, presents the findings of the thesis, and leads a discussion with questions.

Publications

Conference Papers

- [1] Abrar Anwar, John Welsh, Joydeep Biswas, Soha Pouya, and Yan Chang. “ReMEmbR: Building and Reasoning Over Long-Horizon Spatio-Temporal Memory for Robot Navigation.” In: *2025 IEEE International Conference on Robotics and Automation (ICRA)*. <https://nvidia-ai-iot.github.io/remembr/>. Website; a. 2025. URL: <https://arxiv.org/pdf/2409.13682>.
- [2] Joydeep Biswas, Don Fussell, Peter Stone, Kristin Patterson, Kristen Procko, Lea Sabatini, and Zifan Xu. “The Essentials of AI for Life and Society: An AI Literacy Course for the University Community.” In: *Educational Advances in Artificial Intelligence*. Vol. 39. 28. 2025, pp. 28973–28978. DOI: 10.1609/aaai.v39i28.35166. URL: https://joydeepb.com/Publications/eaai2025_essentials.pdf.
- [3] Rohan Chandra, Haresh Karnan, Negar Mehr, Peter Stone, and Joydeep Biswas. “Multi-Agent Inverse Reinforcement Learning in Real World Unstructured Pedestrian Crowds.” In: *Intelligent Robots and Systems (IROS), IEEE/RSJ International Conference on*. 2025. URL: https://joydeepb.com/Publications/iros2025_multi.pdf.
- [4] Rohit Dwivedula, Sadanand Modak, Aditya Akella, Joydeep Biswas, Daehyeok Kim, and Christopher J. Rossbach. “ConfigBot: Adaptive Resource Allocation for Robot Applications in Dynamic Environments.” In: *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2025, pp. 10792–10799. DOI: 10.1109/IROS60139.2025.11245932. URL: https://joydeepb.com/Publications/iros2025_configbot.pdf.
- [5] Eric Hsiung, Joydeep Biswas, and Swarat Chaudhuri. “Automata Learning from Preference and Equivalence Queries.” In: *Proceedings of the 37th International Conference on Computer Aided Verification (CAV 2025)*. Vol. 15934. Springer, 2025, pp. 104–126. DOI: 10.1007/978-3-031-98685-7_5. URL: https://joydeepb.com/Publications/cav2025_remap.pdf.
- [6] Cheng-Chun Hsu, Bowen Wen, Jie Xu, Yashraj Narang, Xiaolong Wang, Yuke Zhu, Joydeep Biswas, and Stan Birchfield. “SPOT: SE(3) Pose Trajectory Diffusion for Object-Centric Manipulation.” In: *Proceedings of the 2025 IEEE International Conference on Robotics and Automation (ICRA)*. 2025. URL: <https://arxiv.org/abs/2411.00965>.
- [7] Zichao Hu, Chen Tang, Michael Joseph Munje, Yifeng Zhu, Alex Liu, Shuijing Liu, Garrett Warnell, Peter Stone, and Joydeep Biswas. “ComposableNav: Instruction-Following Navigation in Dynamic Environments via Composable Diffusion.” In: *Conference on Robot Learning (CoRL)*. 2025.
- [8] Dongmyeong Lee, Jesse Quattrociochi, Christian Ellis, Rwik Rana, Amanda Adkins, Adam Uccello, Garrett Warnell, and Joydeep Biswas. “BEV-Patch-PF: Particle Filtering with BEV-Aerial Feature Matching for Off-Road Geo-Localization.” In: *RSS 2025 Workshop on Resilient Off-road Autonomous Robotics (ROAR)*. Poster presentation. 2025. URL: <https://openreview.net/forum?id=BEV-Patch-PF>.
- [9] Wei Liu, Huihua Zhao, Chenran Li, Joydeep Biswas, Billy Okal, Pulkit Goyal, Soha Pouya, and Yan Chang. “X-Mobility: End-To-End Generalizable Navigation via World Modeling.” In: *2025 IEEE International Conference on Robotics and Automation (ICRA)*. <https://nvlabs.github.io/X-MOBILITY/>. Website; a. 2025. URL: <https://arxiv.org/pdf/2410.17491>.
- [10] Sadanand Modak, Noah Tobias Patton, Isil Dillig, and Joydeep Biswas. “SYNAPSE: SYmbolic Neural-Aided Preference Synthesis Engine.” In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 39. 26. 2025, pp. 27529–27537. DOI: 10.1609/aaai.v39i26.34965. URL: https://joydeepb.com/Publications/aaai2025_synapse.pdf.

- [11] Rwik Rana, Jesse Quattrociocchi, Dongmyeong Lee, Christian Ellis, Amanda Adkins, Adam Uccello, Garrett Warnell, and Joydeep Biswas. “OverSeeC: Open-Vocabulary CostMap Generation from Satellite Images and Natural Language.” In: *RSS 2025 Workshop on Resilient Off-road Autonomous Robotics (ROAR)*. Oral presentation. 2025. URL: <https://openreview.net/forum?id=OverSeeC>.
- [12] Arthur Zhang, Harshit Sikchi, Amy Zhang, and Joydeep Biswas. “CREStE: Scalable Mapless Navigation with Internet Scale Priors and Counterfactual Guidance.” In: *Proceedings of Robotics: Science and Systems*. 2025. URL: https://joydeepb.com/Publications/rss2025_creste.pdf.
- [13] Joydeep Biswas. “Teaching Robots to ”Get It Right”.” In: *Proceedings of the 37th International FLAIRS Conference*. 2024. DOI: 10.32473/flairs.37.1.135854. URL: https://joydeepb.com/Publications/flairs2024_teaching.pdf.
- [14] Hochul Hwang, Hee-Tae Jung, Nicholas A Giudice, Joydeep Biswas, Sunghoon Ivan Lee, and Donghyun Kim. “Towards Robotic Companions: Understanding Handler-Guide Dog Interactions for Informed Guide Dog Robot Design.” In: *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. New York, NY, USA: Association for Computing Machinery, 2024. ISBN: 9798400703300. DOI: 10.1145/3613904.3642181. URL: <https://doi.org/10.1145/3613904.3642181>.
- [15] Haresh Karnan, Elvin Yang, Garrett Warnell, Joydeep Biswas, and Peter Stone. “Wait, That Feels Familiar: Learning to Extrapolate Human Preferences for Preference Aligned Path Planning.” In: *2024 IEEE International Conference on Robotics and Automation (ICRA)*. 2024. DOI: 10.1109/ICRA57147.2024.10611475. URL: https://joydeepb.com/Publications/icra2024_wait.pdf.
- [16] Yung-Chi Kung, Arthur Zhang, Junmin Wang, and Joydeep Biswas. “Looking Inside Out: Anticipating Driver Intent from Videos.” In: *2024 IEEE International Conference on Robotics and Automation (ICRA)*. 2024. DOI: 10.1109/ICRA57147.2024.10610257. URL: https://joydeepb.com/Publications/icra2024_looking.pdf.
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